Quartiles, Box Plots and Outliers

Data Analysis

2026

Preliminary Mathematics Standard

Outline

Quartiles

Interquartile Range

Five-Number Summary and Box Plots

Clusters and Outliers

Homework

Table of Contents

Quartiles

For a more detailed description of the spread, we divide the dataset into smaller parts.

Definition
Percentiles: divide the dataset into 100 parts. To score in the 90th percentile indicates 90% of scores were at or below your score — you scored in the top 10%.

Deciles: divide the dataset into 10 parts.

Quartiles

Definition
First Quartile Q_1 (lower quartile): The 25th percentile. 25% of scores are at or below this value.

Second Quartile Q_2 (median): The 50th percentile. The middle value.

Third Quartile Q_3 (upper quartile): The 75th percentile. 75% of scores are at or below this value.

Calculating Quartiles — Odd Number of Scores

Example
Calculate Q_1, Q_2 and Q_3 for: $1 \quad 2 \quad 2 \quad 3 \quad 5 \quad 7 \quad 8 \quad 9 \quad 10$

1. Order the data (already ordered)
2. $Q_2 =$ median $= 5$ (5th value; omit it)
3. $Q_1 =$ median of left half $\{1, 2, 2, 3\} = 2$
4. $Q_3 =$ median of right half $\{7, 8, 9, 10\} = 8.5$
5. IQR $= 8.5 - 2 = 6.5$

Calculating Quartiles — Even Number of Scores

Example
Calculate Q_1, Q_2 and Q_3 for: $6 \quad 7 \quad 8 \quad 9 \quad 9 \quad 10 \quad 10 \quad 11 \quad 13 \quad 15$

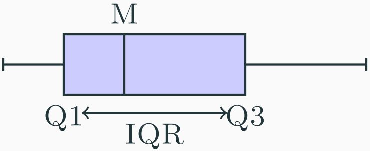
1. Order the data (already ordered)
2. $Q_2 =$ average of 5th and 6th values $= \dfrac{9+10}{2} = 9.5$
3. Split into two halves of 5: $\{6, 7, 8, 9, 9\}$ and $\{10, 10, 11, 13, 15\}$
4. $Q_1 = 8, \quad Q_3 = 11$
5. IQR $= 11 - 8 = 3$

Table of Contents

The interquartile range is a measure of spread covering the middle 50% of the data.

Definition

$$\mathsf{IQR} = Q_3 - Q_1$$



Range vs IQR as Measures of Spread

Range	IQR
Uses only maximum and minimum	Uses the middle 50% of data
Simple to calculate	Slightly more involved
Heavily affected by outliers	Resistant to outliers
Gives full extent of data	Ignores extremes
Can be misleading with skewed data	More reliable for skewed data

General rule: Use IQR when outliers are present or data is skewed. Use range for a quick overview of the full spread.

Table of Contents

Five-Number Summary

Definition

$$\text{Minimum}, \quad Q_1, \quad \text{Median}, \quad Q_3, \quad \text{Maximum}$$

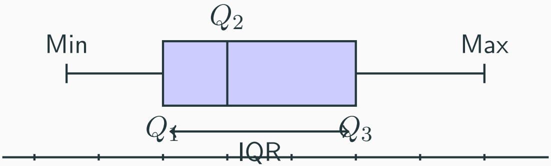
The five-number summary provides a compact description of both the centre and spread of a dataset.

Example
Dataset: $1 \ \ 2 \ \ 2 \ \ 3 \ \ 5 \ \ 7 \ \ 8 \ \ 9 \ \ 10$

$$\text{Min} = 1, \quad Q_1 = 2, \quad Q_2 = 5, \quad Q_3 = 8.5, \quad \text{Max} = 10$$

Box-and-Whisker Plots

A **box plot** is a graphical display of the five-number summary.



- The **box** spans Q_1 to Q_3
- The **line** inside the box marks the median
- The **whiskers** extend to the minimum and maximum (excluding outliers)

Parallel Box Plots — Example

Example
The results of a major exam for two classes are given below.

Class 1: 64 66 68 70 75 77 78 79 81 82 82 82 85 89 89 90 93 93 93 94 94

Class 2: 23 37 60 64 65 66 70 71 72 72 75 75 76 76 78 80 80 90 92 95 99

Construct a parallel box plot and compare the two distributions.

Comparing Distributions Using Parallel Box Plots

When interpreting parallel box plots, comment on:

- **Centre** — compare medians: which group scored higher on average?
- **Spread** — compare IQR and range: which group is more consistent?
- **Shape** — is the data symmetric, positively or negatively skewed?
- **Outliers** — are there any unusual values in either dataset?

Skewness from a box plot:

- Median closer to $Q_1 \Rightarrow$ positively skewed (tail to the right)
- Median closer to $Q_3 \Rightarrow$ negatively skewed (tail to the left)
- Median in centre of box $\Rightarrow$ approximately symmetric

Quartiles from Histograms and Dot Plots

Quartiles can be determined from **histograms** and **dot plots** by counting data values.

Steps:

1. Count the total number of values n
2. Locate the median position: $\dfrac{n+1}{2}$
3. Locate Q_1: median of the lower half
4. Locate Q_3: median of the upper half
5. Read off the minimum and maximum to complete the five-number summary
6. Draw the box plot from the five-number summary

Example
A dot plot shows values: $2 \quad 2 \quad 3 \quad 4 \quad 4 \quad 5 \quad 6 \quad 6 \quad 7 \quad 8$

$n = 10, \quad \text{Min} = 2, \quad Q_1 = 3.5, \quad Q_2 = 4.5, \quad Q_3 = 6.5, \quad \text{Max} = 8$

Interpreting Box Plots

Box plots allow us to make **conclusions and inferences** about a dataset.

Key questions to ask:

- What is the median? What does it tell us about a typical value?
- How large is the IQR? Is the data consistent or variable?
- Are the whiskers equal in length? Is the data symmetric?
- Are there any outliers? What might explain them?
- In a parallel box plot, which group performed better or more consistently?

Example

A box plot for exam scores has: Min $= 42$, $Q_1 = 58$, Median $= 71$, $Q_3 = 83$, Max $= 96$.

The IQR $= 25$ suggests moderate spread. The median is slightly closer to Q_3, indicating mild negative skew — more students scored in the upper range.

Table of Contents

Clusters, Gaps and Outliers

Definition
Cluster: a group of data values that are close together, separated from the rest.

Gap: an interval where no data values occur.

Outlier: a value that lies at an abnormal distance from the rest of the data.

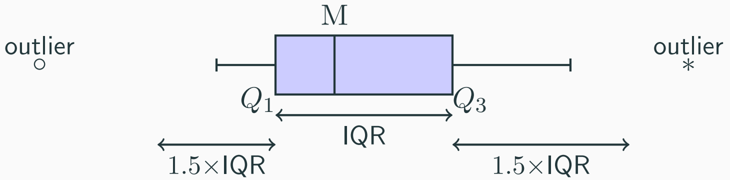
These features should always be interpreted **in context**:

- A cluster may indicate a subgroup within the population
- A gap may suggest two distinct groups (e.g. bimodal distribution)
- An outlier may be a data entry error, or a genuinely unusual observation

Formally Identifying Outliers

Definition
A data value is an **outlier** if:

$$x < Q_1 - 1.5 \times \text{IQR} \qquad \text{or} \qquad x > Q_3 + 1.5 \times \text{IQR}$$



Identifying Outliers — Example

Example
Dataset: $3 \quad 5 \quad 7 \quad 8 \quad 9 \quad 10 \quad 10 \quad 12 \quad 35$

$Q_1 = 6, \quad Q_3 = 11, \quad \text{IQR} = 5$

Lower fence: $6 - 1.5 \times 5 = -1.5$
Upper fence: $11 + 1.5 \times 5 = 18.5$

Since $35 > 18.5$, the value 35 is an **outlier**.

On a box plot, the whisker extends to 12 (the largest non-outlier) and 35 is plotted as a separate point.

Impact of Outliers on Measures of Centre and Spread

Outliers can significantly distort certain statistics.

Statistic	**Effect of outlier**	**Resistant?**
Mean	Pulled toward the outlier	No
Median	Minimally affected	Yes
Mode	Usually unaffected	Yes
Range	Increases significantly	No
IQR	Minimally affected	Yes
σ	Increases significantly	No

Impact of Outliers — Example

Example

Dataset **without** outlier: $5,\ 6,\ 7,\ 8,\ 9,\ 10,\ 11$

$$\bar{x} = 8, \quad \text{Median} = 8, \quad \text{Range} = 6$$

Dataset **with** outlier: $5,\ 6,\ 7,\ 8,\ 9,\ 10,\ 11,\ 45$

$$\bar{x} = 12.6, \quad \text{Median} = 8.5, \quad \text{Range} = 40$$

The mean increased by 4.6 and the range increased by 34, while the median barely changed. This illustrates why the **median and IQR** are preferred when outliers are present.

Outliers in Context

When an outlier is identified, always consider:

- **Data error?** — recording or measurement mistake; should be corrected or removed
- **Genuine observation?** — a real but unusual value; should be kept and noted
- **What does it mean?** — e.g. one student scored 98 when the class averaged 60; this may indicate exceptional ability or different preparation

If an outlier is excluded from analysis, its existence must always be **acknowledged and explained**.

Table of Contents

Homework

Complete all questions from *Data Analysis — Quartiles, Box Plots and Outliers*.